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Special Collection:
Integrating In Situ, Remote Sensing, And Physically Based Modeling Approaches to Understand Global Freshwater Ice Dynamics

Key Points:

- A lightweight drone-borne GPR system is developed for high-resolution ice monitoring
- An algorithm that combines energy ratio and cross-correlation analysis is proposed to automatically estimate the ice thickness
- The results tested in three reservoirs in Tibet and Changchun, China, validate the effectiveness of our GPR system and proposed method

Supporting Information:

Supporting Information may be found in the online version of this article.

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A Drone-Borne GPR System for Lake and River Ice Thickness Monitoring

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Abstract Monitoring ice on water is critical to study global and regional environmental change. However, accurate ice thickness mapping over a water body like a lake is still challenging. In this study, we first develop a drone-borne ground penetrating radar (GPR) system for ice thickness monitoring. This GPR system comprises a micro vector network analyzer, a Vivaldi antenna, a GNSS module, a microcomputer, and an independent power supply. The whole system weighs 1.6 kg, making it easy to mount on a drone. Then, a combined algorithm based on energy ratio and cross-correlation analysis is proposed to automatically extract the time delays between the ice surface and bottom reflections, from which the ice thickness is calculated. Finally, laboratory and field tests were conducted in our laboratory and at three reservoirs in China. Results verified that the GPR system could accurately produce high-resolution ice maps with thicknesses from a few centimeters to ∼one meter. Drilling validated that the relative error of ice thickness estimation is less than 2%. Our findings demonstrate that the developed drone-borne GPR system provides an efficient and accurate method for monitoring ice thickness over lakes, rivers, and reservoirs.

1. Introduction

Climate change has gained a substantial amount of attention worldwide. Ice on lakes and rivers presents the energy exchange between the water body and atmosphere, making it a hot research topic to study lake ice dynamics and the impact of global warming on local or global scales (Culpepper et al., 2025). Additionally, a thick ice layer on lakes and rivers in winter is an important place that accommodates transportation, fishing, recreation, and other human activities (Culpepper et al., 2024; Sharma et al., 2019; Weyhenmeyer et al., 2022). However, recent studies show that the lake ice thickness in winter is decreasing (Basu et al., 2024; Hampton et al., 2024; X. Li et al., 2022). These ice-related activities are threatened by the reduced ice thickness caused by anthropogenically induced climate change (A. M. W. Newton & Mullan, 2021). Meanwhile, regions such as Europe, North America, and Tibet in China, which have abundant water resources, are planning to build or have built many pumped-storage power stations (International Hydropower Association, 2024). The thickness and distribution of ice are important for the related hydraulic structures to avoid ice damage. Therefore, fast and accurate ice thickness monitoring is essential to guarantee the safety of ice-related activities on lakes and rivers, and can provide key information for the construction and operation of hydraulic structures.

Methods for ice thickness detection can be divided into three categories: direct measurement, remote sensing methods, and geophysical methods. A direct method by drilling or a hot-wire thickness gauge can provide reliable and high-accuracy results, but only on a point scale (Perovich et al., 2003). These manual methods are labor- and time-consuming, thus can only be conducted when the ice is thick enough and stable. In contrast, remote sensing and geophysical methods can yield a non-destructive measurement or monitoring of ice thickness. Remote sensing technologies, such as synthetic aperture radar (SAR) and surface-based scatterometry, have been applied to achieve efficient monitoring of ice on seas, lakes, and in polar regions (Duguay et al., 2015; Ferguson & Gunn, 2022; Gunn et al., 2015; Leconte et al., 2009). However, these remote sensing technologies are rarely applied to monitor the ice thickness in small and medium size water bodies due to the limitations of horizontal resolution, repetitive observation capacity, and anti-interference ability. The electromagnetic induction (EMI) method is a geophysical method and has been widely applied to quickly measure the sea ice thickness based on the conductivity contrast between sea ice and water (Hunkeler et al., 2016). But the accuracy of the EMI method will decrease when the ice layer is not flat and snow exists on the ice. The upward-looking sonar method uses the sonar

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instrument onboard a submarine or autonomous underwater vehicle to measure the ice thickness. This method is unaffected by weather but is expensive and has blind zones caused by water conditions (Behrendt et al., 2015; Brown & Duguay, 2011).

Ground penetrating radar (GPR) is a near-surface geophysics method that can detect subsurface structures by transmitting and receiving electromagnetic waves (Brandt et al., 2007; Daniels, 2004; J. Li et al., 2025; Liu et al., 2022; Wang et al., 2024; Webb et al., 2018). Due to the low-attenuation nature of electromagnetic waves propagating in ice, GPR has proven its capacity for high-resolution delineation of the ice layer. Several GPR systems operating from tens to thousands of megahertz have been developed to detect ice in seas, lakes, and rivers (Holt et al., 2009; Liu et al., 2014; Tang et al., 2015). Normally, GPR systems are used to map the ice thicknesses in a survey region or monitor the change in ice thickness at a fixed observation point (Fu et al., 2018; Z. Li et al., 2023; Richards et al., 2023; Wei et al., 2022). However, a significant challenge of traditional GPR in ice monitoring is that the investigation must be performed during the frozen period to keep operators and equipment safe. For the large-scale monitoring of ice thickness in polar and high mountainous regions, GPR can be mounted on a helicopter or fixed-wing aircraft to realize remote detection without sacrificing the resolution (Mouginot et al., 2014; Rutishauser et al., 2016; Santin et al., 2019). However, the applications of a helicopter and fixed-wing aircraft are too expensive to investigate the ice conditions over lakes or rivers.

In the last decade, unpiloted aerial vehicles (UAVs), commonly known as drones, have gradually become a popular platform for geophysical investigations because of their low cost, high flexibility, and easy operation. Drone-borne GPR can efficiently scan dangerous or difficult-to-access areas (Acharya et al., 2021; Chandra & Tanzi, 2018; López et al., 2022; Noviello et al., 2022). Several kinds of drone-borne GPR systems with operating frequencies approximately between 500 and 6,000 MHz are introduced to find possible landmines buried in soils up to ∼25 cm deep (Colorado et al., 2017; García-Fernández et al., 2018; Šipoš & Gleich, 2020). Another application of drone-borne GPR is high-resolution soil moisture mapping (Cheng et al., 2023; Ludeno et al., 2018; Wu et al., 2019). Drone-borne radar systems have also been applied to measure ice and snow. Generally, high-frequency radar systems (≥450 MHz) are suitable for high-resolution investigations of ice and snow on lakes and rivers (Pomerleau et al., 2020; Tan et al., 2017; Valence et al., 2022; Yildiz et al., 2021). While low-frequency radar systems (≤80 MHz) are mainly intended for glacier detection in complex environments (Keshmiri et al., 2017; Lodigiani et al., 2022; Ruols et al., 2023; Teisberg et al., 2022). High spatial and temporal resolution monitoring using drone-borne GPR can provide the ice thickness distribution on lakes and rivers, which is crucial for the accurate security assessment of ice-related activities and hydraulic structures. Pomerleau et al. (2020) installed a commercial radar module on a drone and successfully applied the radar system to measure the ice thicknesses on a frozen lake. However, the used commercial radar module is not specifically designed for ice detection, which may lead to larger errors in their measurements. Therefore, there is an urgent need for a new instrument specifically tailored for ice thickness monitoring.

In previous studies, the Vector Network Analyzer (VNA), as a kind of high-precision radio-frequency instrument, has been applied to build radar systems (Lodigiani et al., 2022; Wu et al., 2019; Šipoš & Gleich, 2020). The VNA serves as both the transmitter and receiver, offering the flexibility to set parameters for a wide range of measurements (Corr et al., 2002; Liu et al., 2022; López et al., 2022). The aim of this study is to develop a high-efficiency and accurate method to map ice thickness in lakes and rivers. Firstly, we develop a drone-borne GPR system consisting of a specially designed Vivaldi antenna, a micro VNA, a GNSS module, a microcomputer, and an independent power source. Then, an algorithm that combines energy ratio and cross-correlation analysis is proposed to automatically estimate the ice thickness. Finally, laboratory experiments and field tests at three reservoirs in China's Tibet and northeast regions were performed to validate the detection capability of the designed GPR system. This study provides a high-efficiency and accurate method to map the lake and river ice thickness with high spatial and temporal resolution. Our work can promote the safety of ice-related activities, the building or maintenance of ice roads, and the construction of hydraulic engineering in cold regions. The rest of this research is organized as follows. Section 2 introduces the developed drone-borne GPR system. The proposed algorithm and experimental results are presented in Sections 3 and 4, respectively. In Section 5, the results are discussed. The conclusions are given in Section 6.

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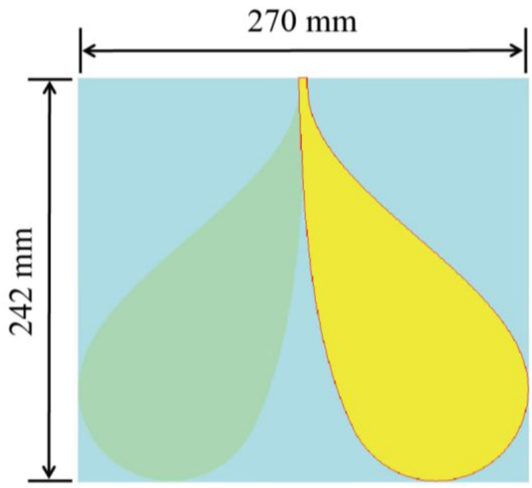

Figure 1. Geometry of the designed Vivaldi antenna.

2. Drone-Borne GPR System

2.1. Antenna Design

A GPR antenna plays a critical role in transmitting and receiving EM waves, significantly impacting the detection capability of a GPR system. The Vivaldi antenna offers several advantages, including good directivity, ultra-wide bandwidth, and lightweight, making it a good choice for a drone-borne GPR system (Liu & Sato, 2013). Therefore, an antipodal Vivaldi antenna is designed for ice monitoring, as illustrated in Figure 1.

Numerical simulations and real tests were conducted to evaluate the designed antenna. In the test, two identical antennas were placed in the boresight direction of each other with a spacing of 2 m (Figure 2a), and the S-parameters were obtained using a VNA. Comparisons between the simulated and measured results, including reflection coefficient, time-domain waveform, and frequency spectrum, are presented in Figures 2b–2d. The measured results agreed well with the simulated ones. The measured return loss and transmission signal show that the effective bandwidth is approximately from ~0.85 to 3.13 GHz with a central frequency of ~2.0 GHz (Figure 2d). Thus, the relative bandwidth is over 110%. According to Jol, 2009; Liu et al., 2014, the range resolution along the radial from the antenna is greater than or equal to one fourth of the half-amplitude width of the pulse envelope. Thus, the range resolution of the Vivaldi antenna is ~2.2 cm in a pure water ice layer (Figure S1 in Supporting Information S1).

2.2. GPR System

As shown in Figure 3, the GPR system consists of a micro VNA, an independently designed Vivaldi antenna, a GNSS module, a microcomputer, and a battery. The VNA works as the transmitter and receiver of a radar system

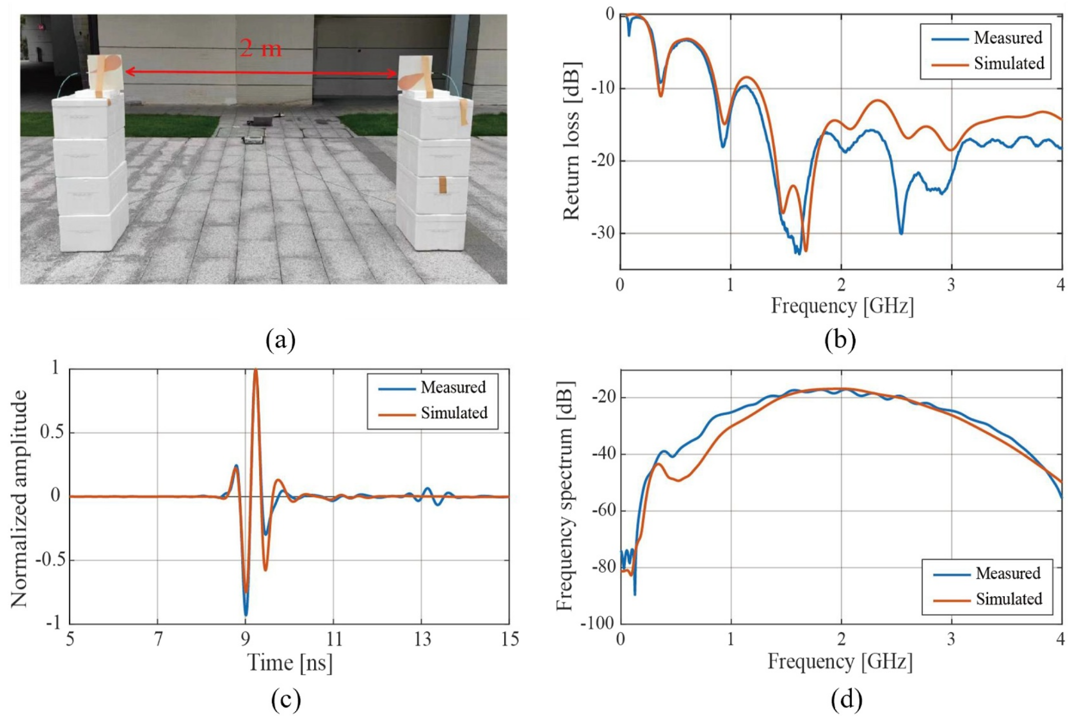

Figure 2. Test of the designed Vivaldi antenna. (a) A photo during the experiment, (b) Return loss, (c) Time-domain transmission signal, and (d) Frequency-domain transmission signal. The reflections at the time of ~13 ns of the measured signal in (c) are caused by ground reflection.

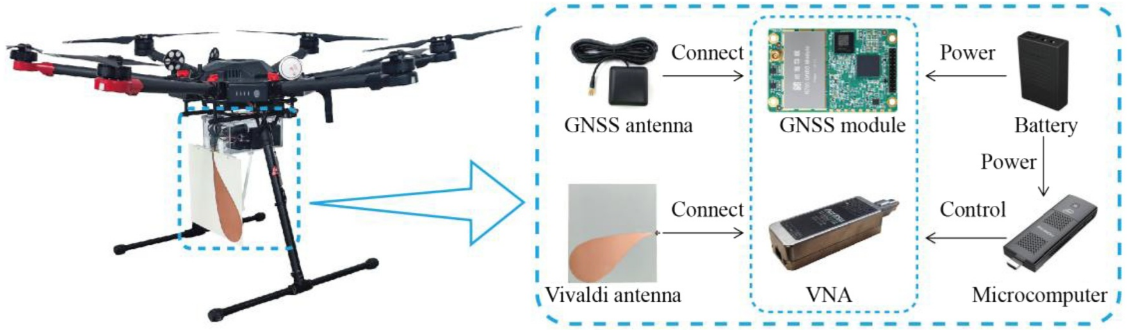

Figure 3. Developed drone-borne GPR system.

in a stepped-frequency continuous-wave mode. To minimize the weight of the GPR system, a single Vivaldi antenna is employed for both transmitting and receiving. The trajectory of the drone-borne GPR system is recorded by a lightweight GNSS module. In real-time kinematic (RTK) mode, the GNSS module can achieve a centimeter-level positioning precision. The data update frequency of GNSS can be freely set as needed. The microcomputer features a quad-core CPU (Intel J4125), 8 GB of memory, 256 GB of hard disk capacity, and runs on Windows 10. The microcomputer controls the VNA and GNSS, including setting parameters, starting and stopping data acquisition, and storing data. A smartphone or laptop is employed to remotely control the microcomputer through a local Wi-Fi network provided by the microcomputer.

All components of the GPR system, except for the Vivaldi antenna, are housed in a waterproof box. This box is securely installed on the bottom of the drone, while the Vivaldi antenna is mounted underneath the box to minimize potential reflections from the drone and its components. The total weight of the GPR system is 1.6 kg, significantly lighter than the drone's maximum load capacity of 6.0 kg. The GPR system is mounted on a DJI M600 Pro drone, which can fly in wind conditions weaker than 8 m/s.

The settings of VNA parameters are optimized for the developed drone-borne GPR system. Generally, the data collection speed is determined by the number of sampling points and intermediate frequency bandwidth (IFBW). These measuring parameters are optimized to balance the detection efficiency and data quality. Under the assumption of flight altitude ≤5 m and ice thickness ≤1 m, the IFBW is set to be 5 kHz, and the operating frequency ranges from 0.02 to 6 GHz with a frequency step of 0.02 GHz. Approximately 12 GPR traces can be collected per second. Usually, the drone's flight speed is controlled to be about 1 m/s. It means that ice thickness can be measured every ∼8 cm in the horizontal direction.

3. Methods

3.1. Preprocessing

To suppress the clutters caused by the drone and the antenna ringing, the raw GPR data is processed by subtracting background data collected at a flight altitude of ∼30 m (Figure S2 in Supporting Information S1). Subsequently, a bandpass filter with a trapezoidal window of $0.1f_c$, $1.4f_c$, $2.0f_c$, and $3.0f_c$ (f_c is the center frequency of the drone-borne GPR system) is applied to the GPR data to enhance the signal-to-noise ratio. Then, the frequency-domain data is converted into the time-domain signal using the inverse Fourier transformation.

3.2. Estimation of Ice Thickness

For a GPR signal, its energy density would significantly increase at the time of first arrival. The energy ratio at a given time is defined as the ratio between the sums of energy in the time windows before and after that time (Zhang et al., 2013), as given by:

$$\mathrm{ER}(t) = \frac{\left[\sum_{t}^{t+L} A^2(\hat{t})\right]^{1/2} + \lambda}{\left[\sum_{t-L}^{t} A^2(\hat{t})\right]^{1/2} + \lambda} \tag{1}$$

where $A(\hat{t})$ represents the amplitude of the radar signal at the time point $\hat{t}$, L is the length of the time window, and λ is a stabilizing factor. When the energy ratio reaches its maximum value at time t_{sfc}, the time point t_{sfc} is the arrival time of the ice surface reflection. For Equation 1, one-tenth of the maximum amplitude value is recommended for λ. However, it is important to note that the energy ratio method is sensitive to the signal-to-clutter ratio (SCR) of the GPR data. As a result, determining the value of time window L requires significant computational effort (Lee et al., 2017).

To optimize the process of ice thickness estimation, we only apply the energy ratio method to extract the first arrival time of a reference trace in the GPR data. Then, the first arrival times of the other traces in the remaining data are obtained through cross-correlation analysis. The cross-correlation between two signals is expressed as follows:

$$R_{mn}(\tau) = \sum_{t_1}^{t_2} A_m(t) \cdot A_n(t + \tau) \tag{2}$$

where A_m is the reference trace, A_n is the test trace, and t represents the time. The maximum value of R_{mn} at the time τ means that the τ is the time delay between the reference trace and the test trace. If using t'_{ref} represents the location of ice surface reflection in the reference trace, the first arrival time of the test trace is $t'_{\mathrm{ref}} + \tau$.

The combination of energy ratio and cross-correlation analysis not only reduces the computation cost but also can provide high-accuracy results of the time delays. A detailed pseudo code, which is performed to extract the first arrival time of ice surface and bottom reflections, is presented in Table S1 in Supporting Information S1. After the extractions of the first arrival time t_{sfc} of ice surface reflections and t_{btm} of ice bottom reflection, the ice thickness h is calculated from the two-way time delay between the ice surface and bottom reflections ($\Delta t = t_{\mathrm{btm}} - t_{\mathrm{sfc}}$), as given by:

$$h = \frac{c\Delta t}{2\sqrt{\varepsilon_{r,\mathrm{ice}}}} \tag{3}$$

where $\varepsilon_{r,\mathrm{ice}}$ is the relative permittivity of ice, and c is the speed of electromagnetic waves in free space. For pure water ice, its relative permittivity is equal to 3.2 (Hilhorst et al., 2001). Additionally, altitude correction is applied to the processed GPR data for better presentation by correcting the first arrival time of ice surface reflections to a flat ground surface. The whole algorithm procedure is shown in Figure 4.

4. Experiments and Results

4.1. Laboratory Experiments

Experiments were conducted to verify the detection capability of the designed Vivaldi antenna in a laboratory environment. The experimental setup is illustrated in Figure 5. The thicknesses of the two ice samples were 5.4 and 45 cm, respectively. To slow down the melting of ice samples, the water used in the pool was pre-frozen in a freezer until it reached a temperature close to 0°C. During the experiments, the trace interval was set to be 5 mm. The data acquisition process of one ice sample was completed within 5 min based on the automatic control of the linear motion module and VNA.

The detection results are shown in Figure 6. For the ice samples with different thicknesses, reflections from the ice surface and bottom were distinguishable. Compared with the surface reflections, the reflections from the ice bottom were complex due to the interference caused by the reflections from the floor and the ice's side faces. The time delays between the ice surface and bottom reflections were extracted using the method proposed in Section 3.2. Then, the ice thicknesses were calculated using Equation 3. The absolute and relative errors were less than 0.9 cm and 3.7% (Table S2 in Supporting Information S1), indicating that the Vivaldi antenna reaches the design requirement for detecting ice thickness ranging from a few to tens of centimeters.

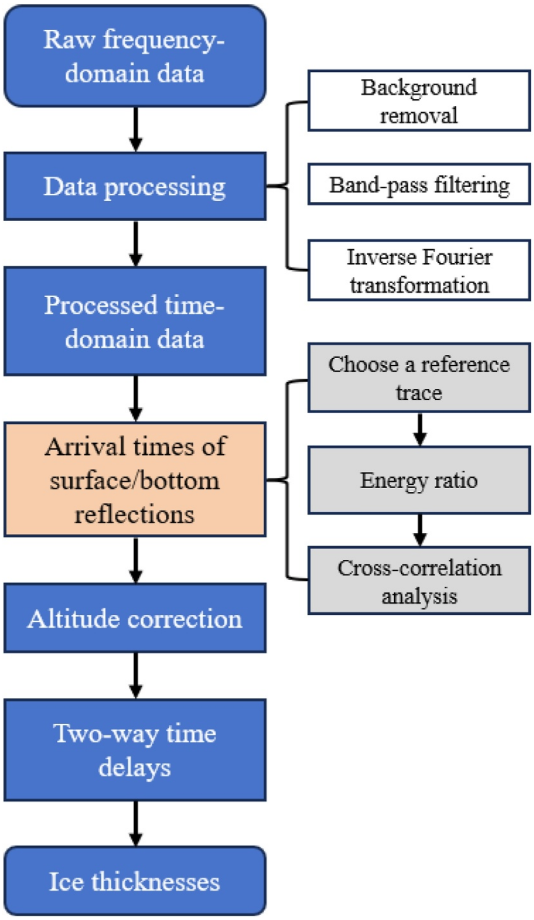

Figure 4. Data processing procedure of ice thickness calculation.

4.2. Field Tests

The first two field tests were conducted at the Pangduo and Baidui reservoirs in Tibet (Figure 7). Tibet covers ∼1.2 million square kilometers and has the most abundant water resources in China. There are over 100 rivers with drainage areas exceeding 2,000 square kilometers. Additionally, Tibet has more than 1,500 lakes, which cover about 24 thousand square kilometers—around 30% of China's total lake area. The Pangduo reservoir is located on the mainstream of the Lhasa River at an elevation of over 4,000 m, whereas the Baidui reservoir is situated at an elevation of over 3,600 m in southeastern Tibet.

The third test was conducted at the Xinlicheng reservoir in Changchun, located in the central part of Northeast China (Figure 7). During the winter of 2024–2025, the average temperature in Changchun was around −12°C. The harsh winters in Northeast China make it a popular destination for a variety of ice-related activities, including ice skating, ice fishing, ice dragon boat racing, ice golf, ice driving, and so on.

Our developed drone-borne GPR system operated in a time-triggered mode, with the flight path manually controlled. The speed of data collection depends on several factors, including the number of sampling points, the intermediate frequency bandwidth (IFBW), and the drone's flight speed. To balance detection efficiency and data quality, the IFBW was set to 5 kHz, and the operating frequency ranged from 0.02 to 6 GHz with a frequency step of 0.02 GHz (i.e., 300 sampling points). During the field tests, the drone's flight speed was maintained at approximately 1 m/s. As a result, about 12 GPR traces were collected per second, resulting in a horizontal resolution of approximately 0.08 m along the survey direction. Ultimately, we obtained ∼1,410 m GPR data and performed six in situ drillings to validate the results in the three field tests.

4.2.1. Field Test 1: Pangduo Reservoir

The first field measurement was performed at the Pangduo reservoir in Tibet on 23 January 2024. The processed GPR profiles along the survey line L1, which is oriented from west to east and ∼24 m in length, are shown in Figure 8. The results demonstrate that the proposed method accurately extracts the first arrivals. Then, the two-travel time delay was calculated based on the extracted first arrival times, allowing for ice thickness estimation. Five points along the survey line were selected

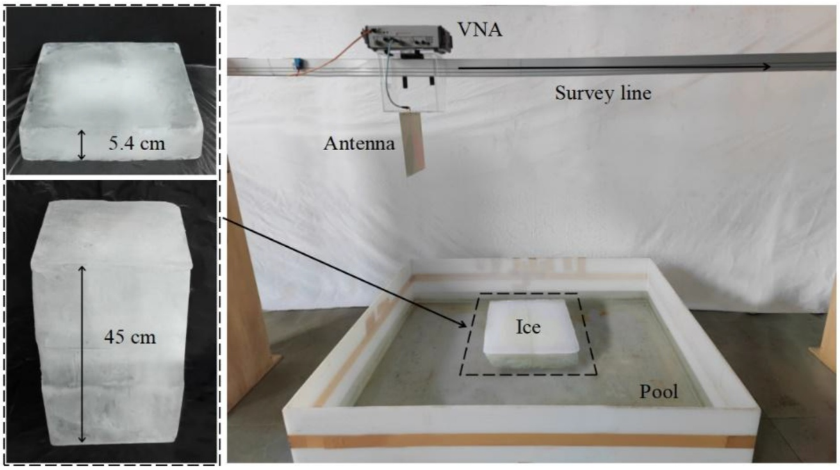

Figure 5. Laboratory experiments for ice thickness detection.

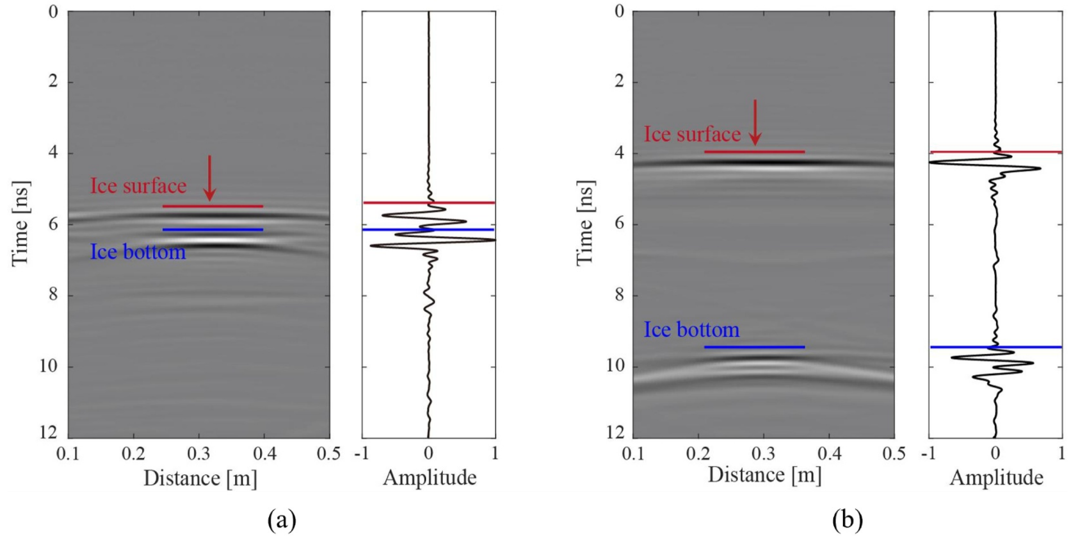

Figure 6. GPR data recorded over the ice samples with a thickness of (a) 5.4 cm and (b) 45.0 cm. The locations of A-scans are marked by the red arrows in the B-scans. The red and blue lines denote the first arrival times of ice surface and bottom reflections.

for drilling to measure the ice thickness using a steel ruler (Figures 8d and 8e). The measured ice thicknesses are compared with the GPR estimates, and the results are given in Table 1. It is revealed that the maximum absolute error of ice thickness estimation is 0.5 cm, showing a relative error of 1.8%. These verifications confirm that the developed drone-borne GPR can detect ice thickness with high resolution and accuracy.

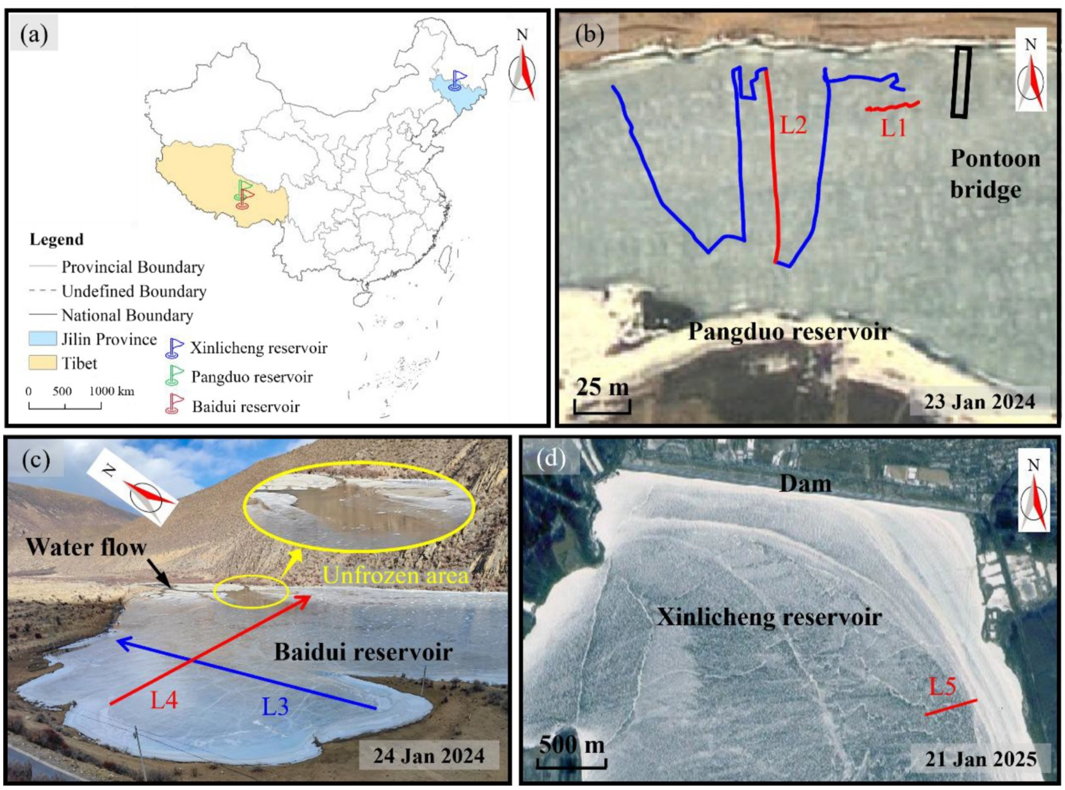

Figure 7. Sites and survey lines of the field tests. (a) Locations of the three field sites, (b) the Pangduo reservoir, (c) the Baidui reservoir, and (d) the Xinlicheng reservoir. The red and blue lines in (b–d) indicate the survey lines.

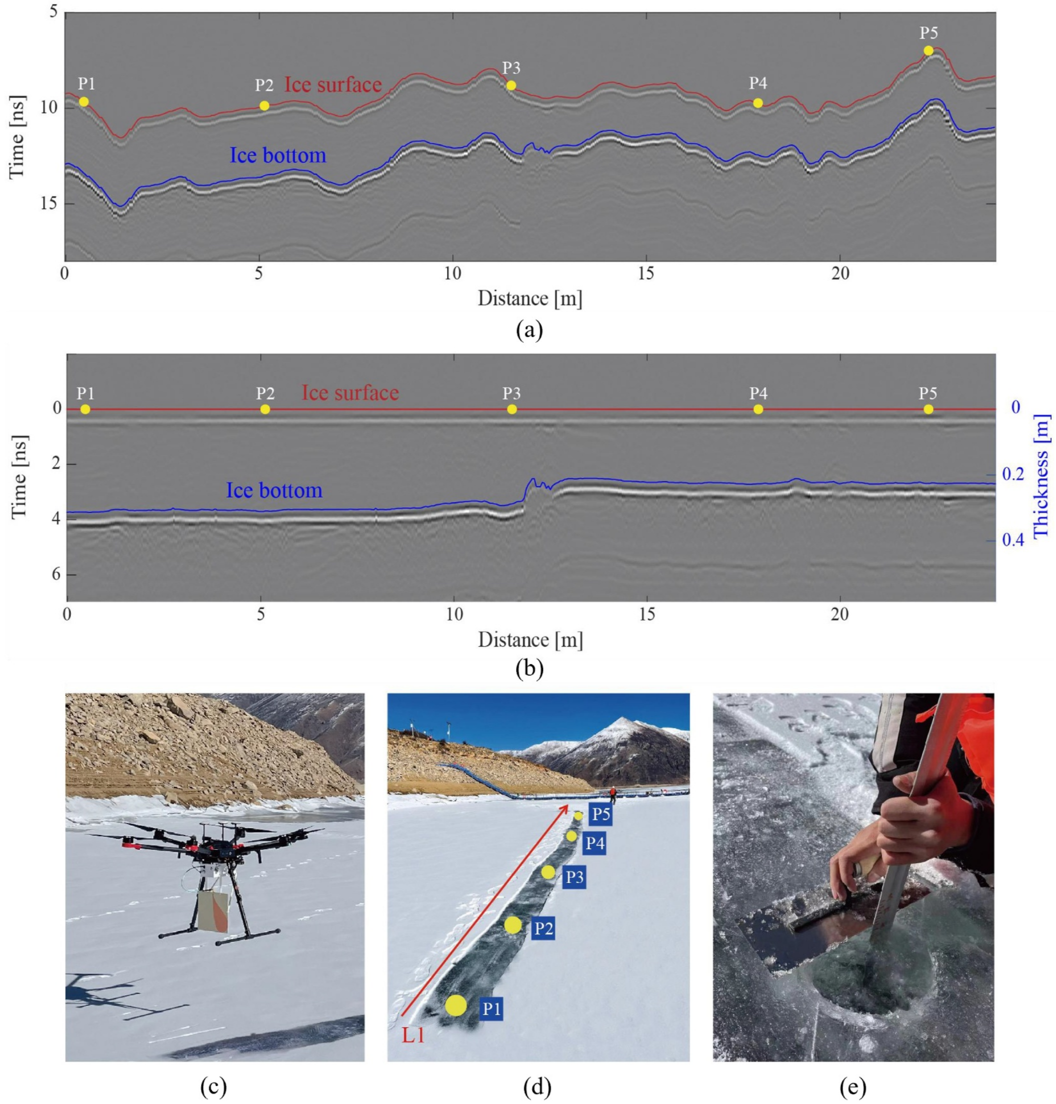

Figure 8. Ice detection along the survey line L1 in the Pangduo reservoir. Processed GPR profiles (a) before and (b) after altitude correction, (c) a field photo, (d) drilling points along the survey line L1, and (e) ice thickness measurement by drilling. The red and blue lines in (a–b) denote the first arrival times of ice surface and bottom reflections, respectively. Note that the removal of surface snow in (c–d) is to ensure that the drone flew along the survey line and provide references to locate the drilling points.

Table 1
Comparison Between the Measured and Estimated Ice Thicknesses Along the Survey Line L1 in the Pangduo Reservoir

Location	Drill measured thickness/cm	Time delay/ns	GPR-estimated thickness/cm	Absolute error/cm	Relative error
P1	30.5	3.67	30.8	0.3	1.0%
P2	30.2	3.64	30.5	0.3	1.0%
P3	27.8	3.38	28.3	0.5	1.8%
P4	22.3	2.68	22.5	0.2	0.9%
P5	22.2	2.68	22.5	0.3	1.2%

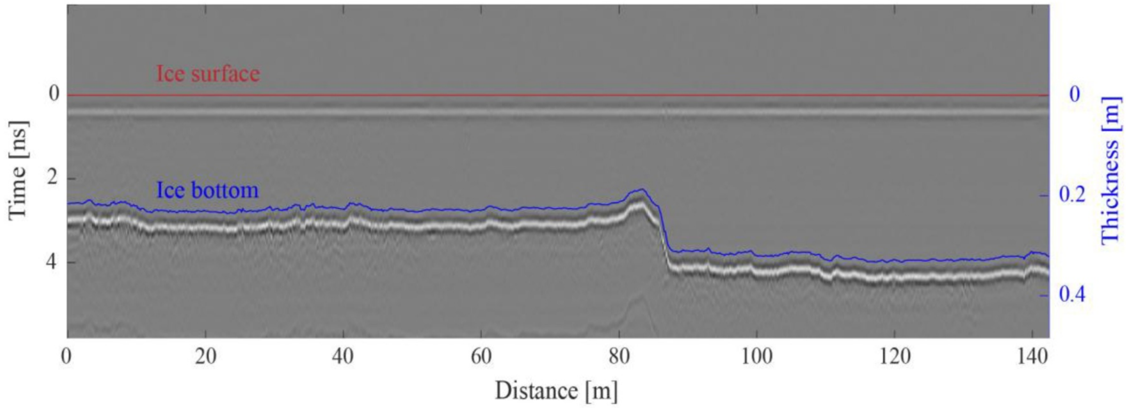

Figure 9. The processed GPR profile after altitude correction along the survey line L2. The red and blue lines denote the first arrival times of ice surface and bottom reflections.

Subsequently, a continuous measurement was conducted along an irregular survey line, covering a detected area of ~14,200 m^2. The results of survey line L2 are presented in Figure 9. The direction of L2 is from south to north and was nearly perpendicular to the survey line L1 (Figure 7b). It was found that the ice thicknesses in the southern region were distinctly thicker than those in the northern region. Then, the ice thicknesses were calculated for all GPR traces and illustrated with the measurement path derived from GNSS data.

The map of ice thicknesses in the survey area is shown in Figure 10. A notable characteristic is the non-homogeneity of the ice thicknesses, with a clear boundary indicating where the thickness changes. The freezing processes in the survey area of Pangduo reservoir are illustrated in Figure 11. Records showed the water surface first froze on the north side, then the ice gradually spread toward the south. The primary reason for the uneven ice was the existence of a pontoon bridge. The bridge reduced the water flow velocity near the north bank, making it easier for the north-side water surface to freeze. As shown in Figure 11, the entire survey area experienced multiple freezing stages before the water surface was completely frozen. Therefore, the reservoir's north part underwent more freezing than the south part, resulting in uneven ice thickness distribution.

4.2.2. Field Test 2: Baidui Reservoir

The second field test was conducted on 24 January 2024, at the Baidui reservoir. Two survey lines, that is, L3 and L4, are 170.3 and 206.1 m, respectively. It is worth noting that there were unfrozen zones near the end of the survey line L4, which might be caused by the warm water from a hot spring nearby.

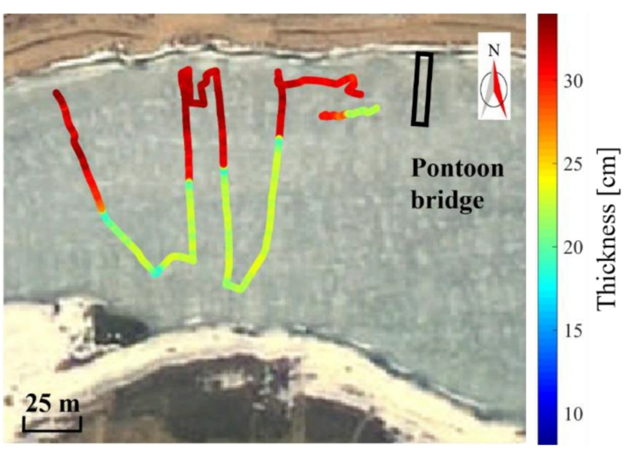

Figure 10. Map of ice thicknesses in the survey area of Pangduo reservoir.

The processed GPR profiles at the Baidui reservoir are shown in Figure 12. It can be found that the time delays between the ice surface and bottom are mainly in the range of ~4.8–5.8 ns, which corresponds to ice thicknesses of ~40–49 cm. The ice thickness at the Baidui reservoir was relatively consistent along the two survey lines except for areas near the shore. At the end of the survey line L4, there are no ice bottom reflections where the water surface was not frozen.

4.2.3. Field Test 3: Xinlicheng Reservoir

The third field test was conducted at the Xinlicheng reservoir in Changchun, China, on 21 January 2025. A survey line with a length of approximately 333 m was set from northeast to southwest (Figure 7d). The processed GPR profile is presented in Figure 13, showing that the ice thicknesses in the Xinlicheng reservoir ranged from 65 to 77 cm, with an average value of ~68 cm. In situ observations by drilling and drone-borne GPR measurement were also performed, and the results present that the absolute error of ice thickness estimation is only 0.4 cm (Figure S3 and Table S3 in Supporting

Figure 11. The field photo of the survey area in Pangduo reservoir at different times. (a) 3 January 2024 and (b) 11 January 2024. Water flows into the reservoir from the right side of the photos.

Information S1). Overall, the ice thicknesses observed in the Xinlicheng reservoir were significantly thicker than those in the Pangduo and Baidui reservoirs, primarily due to the longer and colder days in this region.

5. Discussions

The ice information, for example, maximum ice thickness, minimum ice thickness, and distribution of ice thickness, is not merely important for the safety of ice-related activities, but essential for the design and operation

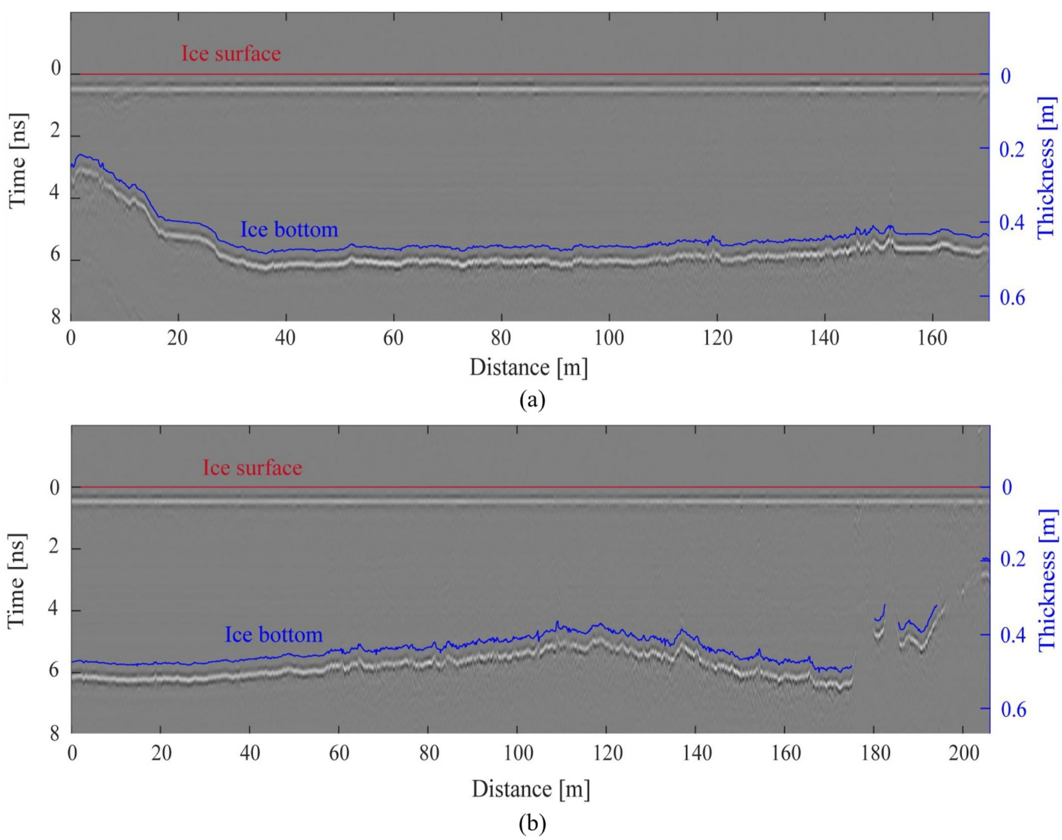

Figure 12. GPR profiles after altitude correction along the survey line L3 (a) and L4 (b) in the Baidui reservoir. The red and blue lines indicate the first arrival times of ice surface and bottom reflections.

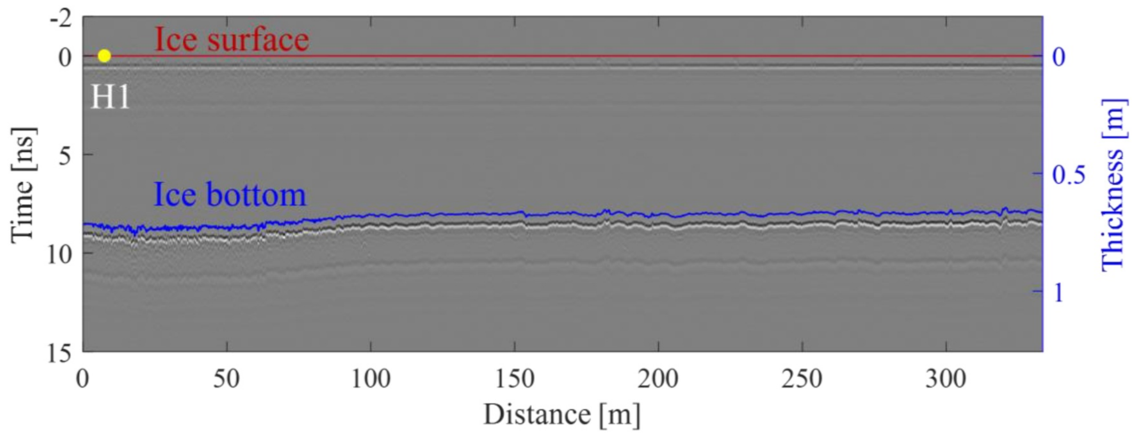

Figure 13. GPR profile after altitude correction along the survey line L5 in the Xinlicheng reservoir. The red and blue lines indicate the first arrival times of ice surface and bottom reflections. The yellow point indicates the drilling hole.

of the hydraulic structures (B. Newton et al., 2024; Qiu et al., 2025). However, the high-resolution ice condition information is usually unavailable due to the lack of field observations. An alternative scenario is using air temperature data to predict the possible range of ice thickness (Lock, 1990). The study performs a comparison between the field tests and temperature-based prediction. According to the empirical formula provided by Water Resources Industry Standards of the People's Republic of China (SL/T 278-2020), the ice thickness h in a reservoir can be estimated through:

$$h = k\left(\sum_{i=1}^{n} |T_i|\right)^{\alpha} \tag{4}$$

where T is the daily average temperature, n is the number of days since the temperature T is stably below 0°C, $k = 1.2 \sim 1.8$ is an empirical coefficient, and $\alpha = 0.50 \sim 0.56$ is an empirical index. Based on Equation 4 and recorded temperature data (Figure S4 in Supporting Information S1), the ice thicknesses in the Pangduo, Baidui, and Xinlicheng reservoirs at the time of the GPR measurements were estimated to be 14.7–30.4 cm, 8.7–17.0 cm, and 28.5–64.0 cm, respectively (Figure 14). The measured ice thicknesses in the Pangduo reservoir fall within the range of the estimates. However, the measured ice thicknesses in the Baidui and Xinlicheng reservoirs are significantly higher than the estimated values. Our findings demonstrate that the empirical formula has limitations when estimating ice thickness in certain reservoirs influenced by various complex environmental factors. The comparison demonstrates that the developed drone-borne GPR system can provide high-resolution ice information for the design and ice damage prevention of hydraulic structures in lakes and rivers.

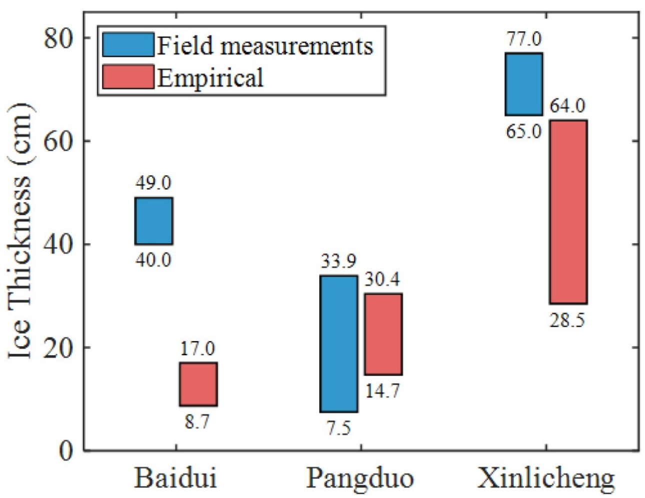

Figure 14. Comparison of ice thickness between the results of drone-borne GPR and empirical formula estimation.

In the field data collections of our drone-borne GPR system, the incidence angle of the radar data can obviously change due to the significant alteration in the drone's flight direction or wind speed. The two-way time delay between the reflections from the ice surface and bottom is generally unaffected by the incidence angle because the electromagnetic wave emitted by the antenna is a plane wave, not a ray. However, the amplitudes of the reflections varied significantly with the change of incidence angle due to the heterogeneity in the antenna's radiation pattern. As a result, inversion methods such as the reflection coefficient method, wave impedance inversion method, and waveform inversion method cannot be directly applied to the drone-borne GPR data for inverting the dielectric property of the ice layer.

6. Conclusions

A drone-borne GPR system has been developed and applied for ice thickness monitoring on three reservoirs in Tibet and Changchun, China. The energy ratio and cross-correlation analysis are combined to automatically calculate the time delay between the ice surface and bottom reflections. Then, the ice thicknesses are obtained based on the ice's relative permittivity of 3.2. The in situ drilling verified the estimated ice thicknesses, with the relative errors being less than 2%. It is concluded that the developed drone-borne GPR system can accurately monitor the ice thickness with high accuracy and resolution, which will be a reliable tool for the safety of ice-related activities and the operation of hydraulic structures in cold regions.

In the future, the research will focus on the inversion of the drone-borne GPR data, including: (a) a high-precision gyroscope will be added to the GPR system to obtain the flight attitude information, which is essential to calculate the incidence angle for the antenna radiation pattern correction; (b) full waveform inversion will be applied to draw the ice permittivity map. Ice permittivity can reveal its moisture content and porosity, providing more comprehensive information to study climate change.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The data used in this paper is accessible at the website of https://doi.org/10.5281/zenodo.14916137.

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References

Acharya, B. S., Bhandari, M., Bandini, F., Pizarro, A., Perks, M., Joshi, D. R., et al. (2021). Unmanned aerial vehicles in hydrology and water management: Applications, challenges, and perspectives. *Water Resources Research*, *57*(11), e2021WR029925. https://doi.org/10.1029/2021WR029925

Basu, A., Culpepper, J., Blagrave, K., & Sharma, S. (2024). Phenological shifts in lake ice cover across the Northern Hemisphere: A glimpse into the past, present, and the future of lake ice phenology. *Water Resources Research*, *60*(9), e2023WR036392. https://doi.org/10.1029/2023WR036392

Behrendt, A., Dierking, W., & Witte, H. (2015). Thermodynamic sea ice growth in the central Weddell Sea, observed in upward-looking sonar data. *Journal of Geophysical Research: Oceans*, *120*(3), 2270–2286. https://doi.org/10.1002/2014JC010408

Brandt, O., Langley, K., Kohler, J., & Hamran, S. E. (2007). Detection of buried ice and sediment layers in permafrost using multi-frequency Ground Penetrating Radar: A case examination on Svalbard. *Remote Sensing of Environment*, *111*(2–3), 212–227. https://doi.org/10.1016/j.rse.2007.03.025

Brown, L. C., & Duguay, C. R. (2011). A comparison of simulated and measured lake ice thickness using a shallow water ice profiler. *Hydrological Processes*, *25*(19), 2932–2941. https://doi.org/10.1002/hyp.8087

Chandra, M., & Tanzi, T. J. (2018). Drone-borne GPR design: Propagation issues. *Comptes Rendus Physique*, *19*(1–2), 72–84. https://doi.org/10.1016/j.crhy.2018.01.002

Cheng, Q., Su, Q., Binley, A., liu, J., Zhang, Z., & Chen, X. (2023). Estimation of surface soil moisture by a multi-elevation UAV-based ground penetrating radar. *Water Resources Research*, *59*(2), e2022WR032621. https://doi.org/10.1029/2022WR032621

Colorado, J., Perez, M., Mondragon, I., Mendez, D., Parra, C., Devia, C., et al. (2017). An integrated aerial system for landmine detection: SDR-based Ground Penetrating Radar onboard an autonomous drone. *Advanced Robotics*, *31*(15), 791–808. https://doi.org/10.1080/01691864.2017.1351393

Corr, H. F., Jenkins, A., Nicholls, K. W., & Doake, C. S. M. (2002). Precise measurement of changes in ice-shelf thickness by phase-sensitive radar to determine basal melt rates. *Geophysical Research Letters*, *29*(8), 73-1–74-4. https://doi.org/10.1029/2001GL014618

Culpepper, J., Jakobsson, E., Weyhenmeyer, G. A., Hampton, S. E., Obertegger, U., Shchapov, K., et al. (2024). Lake ice quality in a warming world. *Nature Reviews Earth & Environment*, *5*(10), 671–685. https://doi.org/10.1038/s43017-024-00590-6

Culpepper, J., Sharma, S., Gunn, G., Magee, M. R., Meyer, M. F., Anderson, E. J., et al. (2025). One-hundred fundamental, open questions to integrate methodological approaches in lake ice research. *Water Resources Research*, *61*(5), e2024WR039042. https://doi.org/10.1029/2024WR039042

Daniels, D. J. (2004). *Ground penetrating radar* (2nd ed.). Institution of Engineering and Technology. London.

Duguay, C. R., Bernier, M., Gauthier, Y., & Kouraev, A. (2015). Remote sensing of lake and river ice. In M. Tedesco (Ed.), *Remote sensing of the cryosphere* (Vol. 12, pp. 273–297). John Wiley & Sons, Ltd. https://doi.org/10.1002/9781118368909

Ferguson, J. E., & Gunn, G. E. (2022). Polarimetric decomposition of microwave-band freshwater ice SAR data: Review, analysis, and future directions. *Remote Sensing of Environment*, *280*, 113176. https://doi.org/10.1016/j.rse.2022.113176

Fu, H., Liu, Z., Guo, X., & Cui, H. (2018). Double-frequency ground penetrating radar for measurement of ice thickness and water depth in rivers and canals: Development, verification and application. *Cold Regions Science and Technology*, *154*, 85–94. https://doi.org/10.1016/j.coldregions.2018.06.017

García-Fernández, M., Álvarez López, Y., Arboleya Arboleya, A., González Valdés, B., Rodríguez Vaqueiro, Y., Las-Heras Andrés, F., & Pino García, A. (2018). Synthetic aperture radar imaging system for landmine detection using a ground penetrating radar on board a unmanned aerial vehicle. *IEEE Access*, *6*, 45100–45112. https://doi.org/10.1109/ACCESS.2018.2863572

Gunn, G. E., Duguay, C. R., Brown, L. C., King, J., Atwood, D., & Kasurak, A. (2015). Freshwater lake ice thickness derived using surface-based X-and Ku-band FMCW scatterometers. *Cold Regions Science and Technology*, *120*, 115–126. https://doi.org/10.1016/j.coldregions.2015.09.012

Hampton, S. E., Powers, S. M., Dugan, H. A., Knoll, L. B., McMeans, B. C., Meyer, M. F., et al. (2024). Environmental and societal consequences of winter ice loss from lakes. *Science*, *386*(6718), 162. https://doi.org/10.1126/science.adl3211

Hilhorst, M. A., Dirksen, C., Kampers, F. W. H., & Feddes, R. A. (2001). Dielectric relaxation of bound water versus soil matric pressure. *Soil Science Society of America Journal*, *65*(2), 311–314. https://doi.org/10.2136/sssaj2001.652311x

Holt, B., Kanagaratnam, P., Gogineni, S. P., Ramasami, V. C., Mahoney, A., & Lytle, V. (2009). Sea ice thickness measurements by ultra-wideband penetrating radar: First results. *Cold Regions Science and Technology*, *55*(1), 33–46. https://doi.org/10.1016/j.coldregions.2008.04.007

Hunkeler, P. A., Hendricks, S., Hoppmann, M., Farquharson, C. G., Kalscheuer, T., Grab, M., et al. (2016). Improved 1D inversions for sea ice thickness and conductivity from electromagnetic induction data: Inclusion of nonlinearities caused by passive bucking. *Geophysics*, *81*(1), WA45–WA58. https://doi.org/10.1190/geo2015-0130.1

International Hydropower Association. (2024). World hydropower outlook. Retrieved from https://acrobat.adobe.com/id/urn:aaid:sc:us:6ba5f8fc-5ad3-4d52-a83c-0931ce5fa119?viewer%21megaVerb=group-discover

Jol, H. M. (2009). *Ground penetrating radar: Theory and applications* (1st ed.). Elsevier Science. Amsterdam, The Netherlands.

Keshmiri, S., Arnold, E., Blevins, A., Ewing, M., Hale, R., Leuschen, C., et al. (2017). Radar ECHO sounding of Russell Glacier at 35 MHz using compact radar systems on small unmanned aerial vehicles. In *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)* (pp. 5900–5903). IEEE. https://doi.org/10.1109/IGARSS.2017.8128352

Leconte, R., Daly, S., Gauthier, Y., Yankielun, N., Bérubé, F., & Bernier, M. (2009). A controlled experiment to retrieve freshwater ice characteristics from an FM-CW radar system. *Cold Regions Science and Technology*, *55*(2), 212–220. https://doi.org/10.1016/j.coldregions.2008.04.003

Lee, M., Byun, J., Kim, D., Choi, J., & Kim, M. (2017). Improved modified energy ratio method using a multi-window approach for accurate arrival picking. *Journal of Applied Geophysics*, *139*, 117–130. https://doi.org/10.1016/j.jappgeo.2017.02.019

Li, J., Liu, H., Meng, X., Duan, D., Lu, H., Zhang, J., et al. (2025). Ancient ocean coastal deposits imaged on Mars. *Proceedings of the National Academy of Sciences*, *122*(9), e2422213122. https://doi.org/10.1073/pnas.2422213122

Li, X., Long, D., Huang, Q., & Zhao, F. (2022). The state and fate of lake ice thickness in the Northern 414 Hemisphere. *Science Bulletin*, *67*(5), 537–546. https://doi.org/10.1016/j.scib.2021.10.015

Li, Z., Li, C., Yang, Y., Zhang, B., Deng, Y., & Li, G. (2023). Physical mechanism and parameterization for correcting radar wave velocity in Yellow River ice with air temperature and ice thickness. *Remote Sensing*, *15*(4), 1121. https://doi.org/10.3390/rs15041121

Liu, H., & Sato, M. (2013). Determination of the phase center position and delay of a Vivaldi antenna. *IEICE Electronics Express*, *10*(21), 20130573. https://doi.org/10.1587/elex.10.20130573

Liu, H., Takahashi, K., & Sato, M. (2014). Measurement of dielectric permittivity and thickness of snow and ice on a brackish lagoon using GPR. *Ieee Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, *7*(3), 820–827. https://doi.org/10.1109/JSTARS.2013.2266792

Liu, H., Zhong, J., Ding, F., Meng, X., Liu, C., & Cui, J. (2022). Detection of early-stage rebar corrosion using a polarimetric ground penetrating radar system. *Construction and Building Materials*, *317*, 125768. https://doi.org/10.1016/j.conbuildmat.2021.125768

Lock, G. S. H. (1990). *The growth and decay of ice*. Cambridge University Press. New York.

Lodigiani, M., Silvestri, L., & Pasian, M. (2022). Glacier monitoring with dual-receiver radar architecture: Preliminary experimental results. In *IGARSS 2022–2022 IEEE International Geoscience and Remote Sensing Symposium* (pp. 3794–3797). https://doi.org/10.1109/IGARSS46834.2022.9883574

López, Y. Á., García-Fernández, M., Álvarez-Narciandi, G., & Andrés, F. L.-H. (2022). Unmanned aerial vehicle-based ground-penetrating radar systems: A review. *IEEE Geoscience and Remote Sensing Magazine*, *10*(2), 66–86. https://doi.org/10.1109/MGRS.2022.3160664

Ludeno, G., Catapano, I., Renga, A., Vetrella, A. R., Fasano, G., & Soldovieri, F. (2018). Assessment of a micro-UAV system for microwave tomography radar imaging. *Remote Sensing of Environment*, *212*, 90–102. https://doi.org/10.1016/j.rse.2018.04.040

Mouginot, J., Rignot, E., Gim, Y., Kirchner, D., & Meur, E. L. (2014). Low-frequency radar sounding of ice in East Antarctica and southern Greenland. *Annals of Glaciology*, *55*(67), 138–146. https://doi.org/10.3189/2014AoG67A089

Newton, A. M. W., & Mullan, D. J. (2021). Climate change and Northern Hemisphere lake and river ice phenology from 1931–2005. *The Cryosphere*, *15*(5), 2211–2234. https://doi.org/10.5194/tc-15-2211-2021

Newton, B., Beltaos, S., & Burrell, B. C. (2024). Ice regimes, ice jams, and a changing hydroclimate, Saint John (Wolastoq) River, New Brunswick, Canada. *Natural Hazards*, *120*(14), 12613–12642. https://doi.org/10.1007/s11069-024-06736-5

Noviello, C., Gennarelli, G., Esposito, G., Ludeno, G., Fasano, G., Capozzoli, L., et al. (2022). An overview on down-looking UVA-based GPR systems. *Remote Sensing*, *14*, 3245. https://doi.org/10.3390/rs14143245

Perovich, D. K., Grenfell, T. C., Richter-Menge, J. A., Light, B., Tucker, W. B., & Eicken, H. (2003). Thin and thinner: Sea ice mass balance measurements during SHEBA. *Journal of Geophysical Research*, *108*(C3), 8050. https://doi.org/10.1029/2001JC001079

Pomerleau, P., Royer, A., Langlois, A., Cliche, P., Courtemanche, B., Madore, J. B., et al. (2020). Low cost and compact FMCW 24 GHz radar applications for snowpack and ice thickness measurements. *Sensors*, *20*(14), 3909. https://doi.org/10.3390/s20143909

Qiu, W. L., Li, K., & Zhao, X. (2025). Study on the deformation and cracking characteristics of bridge-crossing reservoir ice sheet in cold regions. *Cold Regions Science and Technology*, *237*, 104517. https://doi.org/10.1016/j.coldregions.2025.104517

Richards, E., Stuefer, S., Rangel, R. C., Maio, C., Belz, N., & Daanen, R. (2023). An evaluation of GPR monitoring methods on varying river ice conditions: A case study in Alaska. *Cold Regions Science and Technology*, *210*, 103819. https://doi.org/10.1016/j.coldregions.2023.103819

Ruols, B., Baron, L., & Irving, J. (2023). Development of a drone-based ground-penetrating radar system for efficient and safe 3D and 4D surveying of alpine glaciers. *Journal of Glaciology*, *69*(278), 2087–2098. https://doi.org/10.1017/jog.2023.83

Rutishauser, A., Maurer, H., & Bauder, A. (2016). Helicopter-borne ground-penetrating radar investigations on temperate alpine glaciers: A comparison of different systems and their abilities for bedrock mapping. *Geophysics*, *81*(1), WA119–WA129. https://doi.org/10.1190/geo2015-0144.1

Santin, I., Colucci, R. R., Žebre, M., Pavan, M., Cagnati, A., & Forte, E. (2019). Recent evolution of Marmolada glacier (Dolomites, Italy) by means of ground and airborne GPR surveys. *Remote Sensing of Environment*, *235*, 111442. https://doi.org/10.1016/j.rse.2019.111442

Sharma, S., Blagrave, K., Magnuson, J. J., O'Reilly, C. M., Oliver, S., Batt, R. D., et al. (2019). Widespread loss of lake ice around the Northern Hemisphere in a warming world. *Nature Climate Change*, *9*(3), 227–231. https://doi.org/10.1038/s41558-018-0393-5

Šipoš, D., & Gleich, D. (2020). A lightweight and low-power UAV-borne ground penetrating radar design for landmine detection. *Sensors*, *20*(10), 2234. https://doi.org/10.3390/s20082234

Tan, A., Eccleston, K., Platt, I., Woodhead, I., Rack, W., & McCulloch, J. (2017). The design of a UAV mounted snow depth radar: Results of measurements on Antarctic sea ice. In *2017 IEEE Conference on Antenna Measurements & Applications (CAMA)* (pp. 316–319). IEEE. https://doi.org/10.1029/2021WR029925

Tang, X., Sun, B., Guo, J., Liu, X., Cui, X., Zhao, B., & Chen, Y. (2015). A freeze-on ice zone along the Zhongshan–Kunlun ice sheet profile, East Antarctica, by a new ground-based ice-penetrating radar. *Science Bulletin*, *60*(5), 574–576. https://doi.org/10.1007/s11434-015-0732-0

Teisberg, T. O., Schroeder, D. M., Broome, A. L., Lurie, F., & Woo, D. (2022). Development of a UAV-borne pulsed ice-penetrating radar system. In *IGARSS 2022-2022 IEEE International Geoscience and Remote Sensing Symposium* (pp. 7405–7408). https://doi.org/10.1109/IGARSS46834.2022.9883583

Valence, E., Baraer, M., Rosa, E., Barbecot, F., & Monty, C. (2022). Drone-based ground-penetrating radar (GPR) application to snow hydrology. *The Cryosphere*, *16*(9), 3843–3860. https://doi.org/10.5194/tc-16-3843-2022

Wang, X., Liu, H., Meng, X., Cui, J., & Du, Y. (2024). Enhanced imaging of concealed defects behind concrete linings using Residual Channel attention network for rebar clutter suppression. *Automation in Construction*, *166*, 105574. https://doi.org/10.1016/j.autcon.2024.105574

Webb, R. W., Jennings, K. S., Fend, M., & Molotch, N. P. (2018). Combining ground-penetrating radar with terrestrial LiDAR scanning to estimate the spatial distribution of liquid water content in seasonal snowpacks. *Water Resources Research*, *54*(12), 10339–10349. https://doi.org/10.1029/2018WR022680

Wei, Q., Yao, X., Zhang, H., Duan, H., Jin, H., Chen, J., & Cao, J. (2022). Analysis of the variability and influencing factors of ice thickness during the Ablation Period in Qinghai Lake using the GPR ice monitoring system. *Remote Sensing*, *14*(10), 2437. https://doi.org/10.3390/rs14102437

Weyhenmeyer, G. A., Obertegger, U., Rudebeck, H., Jakobsson, E., Jansen, J., Zdorovennova, G., et al. (2022). Towards critical white ice conditions in lakes under global warming. *Nature Communications*, *13*(1), 4974. https://doi.org/10.1038/s41467-022-32633-1

Wu, K., Rodriguez, G. A., Zajc, M., Jacquemin, E., Clément, M., De Coster, A., & Lambot, S. (2019). A new drone-borne GPR for soil moisture mapping. *Remote Sensing of Environment*, *235*, 111456. https://doi.org/10.1016/j.rse.2019.111456

Yildiz, S., Akyurek, Z., & Binley, A. (2021). Quantifying snow water equivalent using terrestrial ground penetrating radar and unmanned aerial vehicle photogrammetry. *Hydrological Processes*, *35*(5), e14190. https://doi.org/10.1002/hyp.14190

Zhang, H. L., Zhu, G. M., & Wang, Y. H. (2013). Automatic microseismic event detection and picking method. *Geophysical and Geochemical Exploration*, *37*(2), 269–273. https://doi.org/10.11720/i.issn.1000-8918.2013.2.17